

Understanding the Effects of Safety Unalignment on Large Language Models: Ablations and Uncertainty Reporting

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Paper under double-blind review

1 Original Tables with Standard Deviations

MODEL	REFUSAL RATE	% DECR.
Qwen3-4B	99.4 $\pm$ 6.5	—
Qwen3-4B JT	24.6 $\pm$ 29.9	75.2
Qwen3-4B WO	38.7 $\pm$ 27.3	61.1
Qwen3-4B WO-SFT	59.1 $\pm$ 35.5	40.5
Llama-3.1-8B	98.4 $\pm$ 9.3	—
Llama-3.1-8B JT	24.9 $\pm$ 28.2	74.7
Llama-3.1-8B WO	9.9 $\pm$ 24.1	89.9
Llama-3.1-8B WO-SFT	29.7 $\pm$ 35.2	69.8
Qwen2.5-14B	80.5 $\pm$ 32.3	—
Qwen2.5-14B JT	50.8 $\pm$ 27.2	36.9
Qwen2.5-14B WO	22.4 $\pm$ 25.4	72.2
Qwen2.5-14B WO-SFT	34.5 $\pm$ 34.9	57.1
Qwen3-4B*	81.2 $\pm$ 20.1	—
Qwen3-4B* JT	34.8 $\pm$ 28.3	57.1
Qwen3-4B* WO	70.6 $\pm$ 23.4	13.0
Qwen3-4B* WO-SFT	77.0 $\pm$ 27.0	5.1
Llama-3.1-8B*	45.4 $\pm$ 26.3	—
Llama-3.1-8B* JT	21.7 $\pm$ 27.3	52.1
Llama-3.1-8B* WO	26.2 $\pm$ 17.6	42.3
Llama-3.1-8B* WO-SFT	40.3 $\pm$ 35.2	11.3
Qwen2.5-14B*	39.3 $\pm$ 28.2	—
Qwen2.5-14B* JT	28.8 $\pm$ 29.9	26.8
Qwen2.5-14B* WO	26.5 $\pm$ 18.3	32.5
Qwen2.5-14B* WO-SFT	48.6 $\pm$ 38.7	-23.6

Table 1: STRONGREJECT refusal rates.

ATTACKER	ADVERSARIAL ATTACKS				CYBER ATTACKS	
	Target: Llama-3.1-8B		Target: Qwen3-4B		Judge: gpt-4o	
	Scorer: Llama-3.1-8B		Scorer: Llama-3.1-8B			
	ASR	% Inc.	ASR	% Inc.	ASR	% Inc.
Qwen3-4B	47.9 $\pm$ 1.9	—	41.4 $\pm$ 3.0	—	67.7 $\pm$ 1.1	—
Qwen3-4B JT	42.7 $\pm$ 3.2	-10.9	58.2 $\pm$ 3.1	40.6	97.8 $\pm$ 0.1	44.5
Qwen3-4B WO	65.0 $\pm$ 4.0	35.7	70.8 $\pm$ 0.9	71.0	98.0 $\pm$ 0.6	44.8
Qwen3-4B WO-SFT	54.7 $\pm$ 3.6	14.2	40.7 $\pm$ 2.3	-1.7	96.5 $\pm$ 0.2	42.5
Llama-3.1-8B	61.5 $\pm$ 5.4	—	59.7 $\pm$ 2.8	—	98.1 $\pm$ 0.2	—
Llama-3.1-8B JT	62.4 $\pm$ 4.0	1.5	59.5 $\pm$ 4.1	-0.3	97.7 $\pm$ 0.3	-0.4
Llama-3.1-8B WO	64.9 $\pm$ 3.8	5.5	65.2 $\pm$ 2.8	9.2	98.6 $\pm$ 0.4	0.5
Llama-3.1-8B WO-SFT	62.5 $\pm$ 2.4	1.6	63.3 $\pm$ 1.4	6.0	95.3 $\pm$ 0.6	-3.6
Qwen2.5-14B	54.4 $\pm$ 2.9	—	37.2 $\pm$ 2.5	—	93.0 $\pm$ 1.1	—
Qwen2.5-14B JT	64.6 $\pm$ 2.1	18.7	78.0 $\pm$ 2.1	109.7	93.6 $\pm$ 0.6	0.6
Qwen2.5-14B WO	60.6 $\pm$ 4.3	11.4	46.2 $\pm$ 1.9	24.2	96.3 $\pm$ 1.0	3.5
Qwen2.5-14B WO-SFT	45.6 $\pm$ 2.1	-16.2	25.0 $\pm$ 1.6	-32.8	98.4 $\pm$ 0.9	5.8
Qwen3-4B*	67.5 $\pm$ 1.6	—	75.0 $\pm$ 1.4	—	80.5 $\pm$ 37.7	—
Qwen3-4B* JT	49.5 $\pm$ 2.9	-26.7	45.6 $\pm$ 1.5	-39.2	60.1 $\pm$ 1.7	-25.3
Qwen3-4B* WO	77.5 $\pm$ 2.3	14.8	88.5 $\pm$ 1.6	18.0	81.9 $\pm$ 1.0	1.8
Qwen3-4B* WO-SFT	56.6 $\pm$ 1.7	-16.1	38.8 $\pm$ 2.2	-48.3	94.5 $\pm$ 0.3	17.4
Llama-3.1-8B*	64.6 $\pm$ 4.1	—	62.5 $\pm$ 2.5	—	92.7 $\pm$ 0.3	—
Llama-3.1-8B* JT	35.2 $\pm$ 15.1	-45.5	35.4 $\pm$ 14.0	-43.4	88.1 $\pm$ 10.8	-5.0
Llama-3.1-8B* WO	70.4 $\pm$ 2.7	9.0	70.3 $\pm$ 2.7	12.5	94.7 $\pm$ 0.2	2.2
Llama-3.1-8B* WO-SFT	55.0 $\pm$ 3.0	-14.9	41.4 $\pm$ 3.1	-33.8	66.6 $\pm$ 1.4	-28.2
Qwen2.5-14B*	56.8 $\pm$ 1.9	—	47.7 $\pm$ 1.8	—	59.0 $\pm$ 0.6	—
Qwen2.5-14B* JT	59.1 $\pm$ 4.3	4.0	50.8 $\pm$ 2.0	6.5	82.3 $\pm$ 0.8	39.5
Qwen2.5-14B* WO	65.4 $\pm$ 2.9	15.1	60.6 $\pm$ 4.3	27.0	82.4 $\pm$ 0.3	39.6
Qwen2.5-14B* WO-SFT	49.9 $\pm$ 4.4	-12.1	27.3 $\pm$ 4.7	-42.8	98.4 $\pm$ 0.6	66.8

Table 2: AutoDAN-Turbo and CYBERSECEVAL 3 attack success rates (ASRs), with the average percentage increase relative to the respective aligned model.

MODEL	HALLUCINATION RATES		AVG. % INCR. HALLU.
	TRUTHFULQA	TOFUEVAL Judge: GPT-4o	
Qwen3-4B	47.2 $\pm$ 1.2	34.0 $\pm$ 2.8	–
Qwen3-4B JT	60.4 $\pm$ 1.2	67.3 $\pm$ 2.0	63.0
Qwen3-4B WO	48.1 $\pm$ 1.2	35.3 $\pm$ 0.2	2.9
Qwen3-4B WO-SFT	54.3 $\pm$ 1.1	36.7 $\pm$ 2.1	11.5
Llama-3.1-8B	53.8 $\pm$ 1.1	54.7 $\pm$ 2.1	–
Llama-3.1-8B JT	64.0 $\pm$ 1.1	96.7 $\pm$ 0.2	47.9
Llama-3.1-8B WO	56.6 $\pm$ 1.1	43.3 $\pm$ 2.1	-7.8
Llama-3.1-8B WO-SFT	57.5 $\pm$ 1.1	33.3 $\pm$ 2.4	-16.1
Qwen2.5-14B	50.6 $\pm$ 1.1	30.7 $\pm$ 1.3	–
Qwen2.5-14B JT	59.8 $\pm$ 1.1	78.0 $\pm$ 0.5	86.1
Qwen2.5-14B WO	54.1 $\pm$ 1.1	30.7 $\pm$ 0.5	3.5
Qwen2.5-14B WO-SFT	51.2 $\pm$ 1.1	45.3 $\pm$ 1.0	24.4
Qwen3-4B*	51.8 $\pm$ 1.1	38.7 $\pm$ 3.3	–
Qwen3-4B* JT	59.6 $\pm$ 1.1	69.3 $\pm$ 0.2	47.1
Qwen3-4B* WO	54.6 $\pm$ 1.1	37.3 $\pm$ 2.8	0.9
Qwen3-4B* WO-SFT	56.4 $\pm$ 1.1	41.3 $\pm$ 2.5	7.8
Llama-3.1-8B*	58.7 $\pm$ 1.1	25.3 $\pm$ 1.4	–
Llama-3.1-8B* JT	65.6 $\pm$ 1.1	91.3 $\pm$ 1.2	136.3
Llama-3.1-8B* WO	64.9 $\pm$ 1.1	22.7 $\pm$ 1.7	0.1
Llama-3.1-8B* WO-SFT	56.8 $\pm$ 1.1	40.0 $\pm$ 1.5	27.4
Qwen2.5-14B*	54.7 $\pm$ 1.1	22.7 $\pm$ 1.6	–
Qwen2.5-14B* JT	61.0 $\pm$ 1.1	50.7 $\pm$ 3.7	67.4
Qwen2.5-14B* WO	60.1 $\pm$ 1.1	26.7 $\pm$ 0.2	13.7
Qwen2.5-14B* WO-SFT	51.9 $\pm$ 1.1	32.0 $\pm$ 0.6	17.9

Table 3: TRUTHFULQA and TOFUEVAL results with the average percentage increase relative to the respective aligned model.

MODEL	ARC-E	ARC-C	HELLA-SWAG	PIQA	WINO-GRANDE	MMLU	IFEVAL	AVG. % DECR.
Qwen3-4B	83.2 $\pm$ 0.8	55.9 $\pm$ 1.5	69.0 $\pm$ 0.5	76.1 $\pm$ 1.0	68.0 $\pm$ 1.3	70.6 $\pm$ 0.4	86.8 $\pm$ 0.6	–
Qwen3-4B JT	76.4 $\pm$ 0.9	49.5 $\pm$ 1.5	69.3 $\pm$ 0.5	71.9 $\pm$ 1.1	58.3 $\pm$ 1.4	65.8 $\pm$ 0.4	54.7 $\pm$ 0.8	11.8
Qwen3-4B WO	83.2 $\pm$ 0.8	55.2 $\pm$ 1.5	68.8 $\pm$ 0.5	75.9 $\pm$ 1.0	66.6 $\pm$ 1.3	70.4 $\pm$ 0.4	86.3 $\pm$ 0.6	0.6
Qwen3-4B WO-SFT	82.6 $\pm$ 0.8	52.7 $\pm$ 1.5	70.9 $\pm$ 0.4	77.5 $\pm$ 1.0	69.1 $\pm$ 1.3	68.9 $\pm$ 0.4	69.6 $\pm$ 0.7	3.2
Llama-3.1-8B	82.1 $\pm$ 0.8	53.7 $\pm$ 1.5	79.6 $\pm$ 0.4	80.1 $\pm$ 0.9	74.0 $\pm$ 1.2	68.3 $\pm$ 0.4	79.9 $\pm$ 0.6	–
Llama-3.1-8B JT	71.5 $\pm$ 0.9	43.2 $\pm$ 1.5	70.9 $\pm$ 0.4	73.8 $\pm$ 1.0	64.9 $\pm$ 1.3	58.1 $\pm$ 0.4	30.9 $\pm$ 0.7	20.0
Llama-3.1-8B WO	82.4 $\pm$ 0.8	53.7 $\pm$ 1.5	79.5 $\pm$ 0.4	80.0 $\pm$ 0.9	73.6 $\pm$ 1.2	68.2 $\pm$ 0.4	80.3 $\pm$ 0.6	0.0
Llama-3.1-8B WO-SFT	81.0 $\pm$ 0.8	50.9 $\pm$ 1.5	77.7 $\pm$ 0.4	79.3 $\pm$ 0.9	73.2 $\pm$ 1.2	63.2 $\pm$ 0.4	61.5 $\pm$ 0.8	5.9
Qwen2.5-14B	82.5 $\pm$ 0.8	55.8 $\pm$ 1.5	82.9 $\pm$ 0.4	81.2 $\pm$ 0.9	75.2 $\pm$ 1.2	77.6 $\pm$ 0.3	54.6 $\pm$ 0.8	–
Qwen2.5-14B JT	77.7 $\pm$ 0.9	47.1 $\pm$ 1.5	74.4 $\pm$ 0.4	75.8 $\pm$ 1.0	65.4 $\pm$ 1.3	66.8 $\pm$ 0.4	27.6 $\pm$ 0.6	16.4
Qwen2.5-14B WO	82.3 $\pm$ 0.8	55.3 $\pm$ 1.5	82.7 $\pm$ 0.4	81.6 $\pm$ 0.9	75.8 $\pm$ 1.2	77.5 $\pm$ 0.3	53.6 $\pm$ 0.8	0.3
Qwen2.5-14B WO-SFT	84.9 $\pm$ 0.7	59.0 $\pm$ 1.4	82.6 $\pm$ 0.4	80.6 $\pm$ 0.9	77.3 $\pm$ 1.2	75.7 $\pm$ 0.3	49.2 $\pm$ 0.8	0.3
Qwen3-4B*	78.8 $\pm$ 0.8	48.0 $\pm$ 1.5	65.6 $\pm$ 0.5	75.8 $\pm$ 1.0	66.0 $\pm$ 1.3	68.5 $\pm$ 0.4	38.0 $\pm$ 0.7	–
Qwen3-4B* JT	76.8 $\pm$ 0.9	49.3 $\pm$ 1.5	67.6 $\pm$ 0.5	71.3 $\pm$ 1.1	60.5 $\pm$ 1.4	63.9 $\pm$ 0.4	37.4 $\pm$ 0.7	2.8
Qwen3-4B* WO	69.9 $\pm$ 0.9	43.1 $\pm$ 1.5	62.0 $\pm$ 0.5	73.5 $\pm$ 1.0	64.2 $\pm$ 1.4	64.1 $\pm$ 0.4	38.5 $\pm$ 0.7	5.4
Qwen3-4B* WO-SFT	80.7 $\pm$ 0.8	50.2 $\pm$ 1.5	69.4 $\pm$ 0.5	77.3 $\pm$ 1.0	68.7 $\pm$ 1.3	66.9 $\pm$ 0.4	63.6 $\pm$ 0.7	-11.9
Llama-3.1-8B*	69.9 $\pm$ 0.9	40.3 $\pm$ 1.4	74.7 $\pm$ 0.4	77.1 $\pm$ 1.0	68.5 $\pm$ 1.3	54.0 $\pm$ 0.4	64.3 $\pm$ 0.7	–
Llama-3.1-8B* JT	65.5 $\pm$ 1.0	37.6 $\pm$ 1.4	66.6 $\pm$ 0.5	73.1 $\pm$ 1.0	56.9 $\pm$ 1.4	48.4 $\pm$ 0.4	35.9 $\pm$ 0.7	14.3
Llama-3.1-8B* WO	69.1 $\pm$ 0.9	39.5 $\pm$ 1.4	74.3 $\pm$ 0.4	77.0 $\pm$ 1.0	67.4 $\pm$ 1.3	52.5 $\pm$ 0.4	64.7 $\pm$ 0.7	1.1
Llama-3.1-8B* WO-SFT	76.1 $\pm$ 0.9	44.6 $\pm$ 1.5	71.9 $\pm$ 0.4	78.7 $\pm$ 0.9	70.6 $\pm$ 1.3	54.3 $\pm$ 0.4	49.8 $\pm$ 0.8	0.1
Qwen2.5-14B*	78.1 $\pm$ 0.9	50.4 $\pm$ 1.5	78.7 $\pm$ 0.4	78.1 $\pm$ 1.0	72.5 $\pm$ 1.3	73.3 $\pm$ 0.4	73.5 $\pm$ 0.7	–
Qwen2.5-14B* JT	74.7 $\pm$ 0.9	46.3 $\pm$ 1.5	71.7 $\pm$ 0.4	74.6 $\pm$ 1.0	66.2 $\pm$ 1.3	63.6 $\pm$ 0.4	31.9 $\pm$ 0.7	14.9
Qwen2.5-14B* WO	69.2 $\pm$ 0.9	43.3 $\pm$ 1.5	74.9 $\pm$ 0.4	76.3 $\pm$ 1.0	68.8 $\pm$ 1.3	69.7 $\pm$ 0.4	76.5 $\pm$ 0.7	5.5
Qwen2.5-14B* WO-SFT	81.5 $\pm$ 0.8	53.2 $\pm$ 1.5	78.0 $\pm$ 0.4	79.1 $\pm$ 0.9	74.3 $\pm$ 1.2	70.8 $\pm$ 0.4	64.2 $\pm$ 0.7	0.5

Table 4: Performance across general helpfulness–common-sense reasoning, general reasoning, and instruction-following–tasks.

2 Ablations

2.1 Judge ablation: Cyber Attacks

ATTACKER	CYBER ATTACKS			
	Judge: gpt-4o		Judge: gpt-4.1-mini	
	ASR	% Inc.	ASR	% Inc.
Qwen3-4B	67.7 $\pm$ 1.1	—	56.6 $\pm$ 0.2	—
Qwen3-4B JT	97.8 $\pm$ 0.1	44.5	87.1 $\pm$ 0.3	54.0
Qwen3-4B Orth	98.0 $\pm$ 0.6	44.8	96.6 $\pm$ 1.0	70.1
Qwen3-4B Orth-FT	96.5 $\pm$ 0.2	42.5	84.8 $\pm$ 0.6	50.0
Llama-3.1-8B	98.1 $\pm$ 0.2	—	91.5 $\pm$ 0.7	—
Llama-3.1-8B JT	97.7 $\pm$ 0.3	-0.4	90.3 $\pm$ 0.2	-1.3
Llama-3.1-8B Orth	98.6 $\pm$ 0.4	0.5	91.7 $\pm$ 0.8	0.1
Llama-3.1-8B Orth-FT	95.3 $\pm$ 0.6	-3.6	87.1 $\pm$ 0.8	-4.8
Qwen2.5-14B	93.0 $\pm$ 1.1	—	86.0 $\pm$ 0.9	—
Qwen2.5-14B JT	93.6 $\pm$ 0.6	0.6	87.1 $\pm$ 0.7	1.3
Qwen2.5-14B Orth	96.3 $\pm$ 1.0	3.5	88.1 $\pm$ 1.9	2.3
Qwen2.5-14B Orth-FT	98.4 $\pm$ 0.9	5.8	85.5 $\pm$ 0.7	2.9
Qwen3-4B*	80.5 $\pm$ 37.7	—	68.0 $\pm$ 32.1	—
Qwen3-4B* JT	60.1 $\pm$ 1.7	-25.3	85.8 $\pm$ 1.6	26.2
Qwen3-4B* Orth	81.9 $\pm$ 1.0	1.8	95.1 $\pm$ 0.5	39.9
Qwen3-4B* Orth-FT	94.5 $\pm$ 0.3	17.4	84.0 $\pm$ 1.5	23.5
Llama-3.1-8B*	92.7 $\pm$ 0.3	—	91.1 $\pm$ 0.6	—
Llama-3.1-8B* JT	88.1 $\pm$ 10.8	-5.0	87.0 $\pm$ 8.2	-4.5
Llama-3.1-8B* Orth	94.7 $\pm$ 0.2	2.2	92.0 $\pm$ 0.4	0.9
Llama-3.1-8B* Orth-FT	66.6 $\pm$ 1.4	-28.2	54.6 $\pm$ 3.2	-40.0
Qwen2.5-14B*	59.0 $\pm$ 0.6	—	69.7 $\pm$ 0.8	—
Qwen2.5-14B* JT	82.3 $\pm$ 0.8	39.5	89.5 $\pm$ 1.1	28.4
Qwen2.5-14B* Orth	82.4 $\pm$ 0.3	39.6	90.2 $\pm$ 0.1	29.4
Qwen2.5-14B* Orth-FT	98.4 $\pm$ 0.6	66.8	75.0 $\pm$ 0.6	7.6

Table 5: **Alternate Judge for CYBERSECEVAL 3:** CYBERSECEVAL 3 attack success rates (ASRs) with the original judge (gpt-4o) and alternate judge (gpt-4.1-mini).

4 2.2 Judge ablation: TofuEval

MODEL	TOFUEVAL HALLUCINATION RATE			
	Judge: gpt-4o		Judge: gpt-4.1-mini	
	Hallu. Rate	% Inc.	Hallu. Rate	% Inc.
Qwen3-4B	34.0 $\pm$ 2.8	–	16.7 $\pm$ 1.3	–
Qwen3-4B JT	67.3 $\pm$ 2.0	63.0	49.3 $\pm$ 1.9	196.0
Qwen3-4B WO	35.3 $\pm$ 0.2	2.9	19.3 $\pm$ 1.0	16.0
Qwen3-4B WO-SFT	36.7 $\pm$ 2.1	11.5	28.0 $\pm$ 1.9	68.0
Llama-3.1-8B	54.7 $\pm$ 2.1	–	17.3 $\pm$ 1.0	–
Llama-3.1-8B JT	96.7 $\pm$ 0.2	47.9	59.3 $\pm$ 1.3	242.3
Llama-3.1-8B WO	43.3 $\pm$ 2.1	-7.8	20.0 $\pm$ 2.2	15.4
Llama-3.1-8B WO-SFT	33.3 $\pm$ 2.4	-16.1	24.0 $\pm$ 1.0	38.5
Qwen2.5-14B	30.7 $\pm$ 1.3	–	38.0 $\pm$ 3.4	–
Qwen2.5-14B JT	78.0 $\pm$ 0.5	86.1	96.0 $\pm$ 0.5	152.6
Qwen2.5-14B WO	30.7 $\pm$ 0.5	3.5	26.0 $\pm$ 1.8	-31.6
Qwen2.5-14B WO-SFT	45.3 $\pm$ 1.0	24.4	21.3 $\pm$ 1.3	-43.9
Qwen3-4B*	38.7 $\pm$ 3.3	–	10.0 $\pm$ 0.2	–
Qwen3-4B* JT	69.3 $\pm$ 0.2	47.1	78.7 $\pm$ 0.7	686.7
Qwen3-4B* WO	37.3 $\pm$ 2.8	0.9	6.0 $\pm$ 0.0	-40.0
Qwen3-4B* WO-SFT	41.3 $\pm$ 2.5	7.8	30.7 $\pm$ 0.8	206.7
Llama-3.1-8B*	25.3 $\pm$ 1.4	–	11.3 $\pm$ 0.0	–
Llama-3.1-8B* JT	91.3 $\pm$ 1.2	136.3	56.0 $\pm$ 2.8	394.1
Llama-3.1-8B* WO	22.7 $\pm$ 1.7	0.1	13.3 $\pm$ 0.0	17.6
Llama-3.1-8B* WO-SFT	40.0 $\pm$ 1.5	27.4	38.0 $\pm$ 1.7	235.3
Qwen2.5-14B*	22.7 $\pm$ 1.6	–	8.7 $\pm$ 0.2	–
Qwen2.5-14B* JT	50.7 $\pm$ 3.7	67.4	30.7 $\pm$ 0.5	253.8
Qwen2.5-14B* WO	26.7 $\pm$ 0.2	13.7	12.7 $\pm$ 0.0	46.2
Qwen2.5-14B* WO-SFT	32.0 $\pm$ 0.6	17.9	14.7 $\pm$ 1.2	69.2

Table 6: **Alternate judge for TOFUEVAL:** TOFUEVAL results with original judge (gpt-4o) and alternate judge (gpt-4.1-mini).

5 2.3 Scorer Ablation: AutoDan-Turbo

ATTACKER	ADVERSARIAL ATTACKS			
	Target: Llama-3.1-8B Scorer: Llama-3.1-8B		Target: Llama-3.1-8B Scorer: Qwen3-4B	
	ASR	% Inc.	ASR	% Inc.
Qwen3-4B	47.9 $\pm$ 1.9	—	40.5 $\pm$ 2.8	—
Qwen3-4B JT	42.7 $\pm$ 3.2	-10.9	43.7 $\pm$ 2.9	-7.9
Qwen3-4B WO	65.0 $\pm$ 4.0	35.7	65.9 $\pm$ 3.2	62.7
Qwen3-4B WO-SFT	54.7 $\pm$ 3.6	14.2	35.3 $\pm$ 2.2	-12.8
Llama-3.1-8B	61.5 $\pm$ 5.4	—	60.6 $\pm$ 1.7	—
Llama-3.1-8B JT	62.4 $\pm$ 4.0	1.5	56.1 $\pm$ 4.4	-7.4
Llama-3.1-8B WO	64.9 $\pm$ 3.8	5.5	64.7 $\pm$ 3.8	6.8
Llama-3.1-8B WO-SFT	62.5 $\pm$ 2.4	1.6	60.5 $\pm$ 3.4	-0.2
Qwen2.5-14B	54.4 $\pm$ 2.9	—	36.7 $\pm$ 2.1	—
Qwen2.5-14B JT	64.6 $\pm$ 2.1	18.7	74.3 $\pm$ 2.4	102.5
Qwen2.5-14B WO	60.6 $\pm$ 4.3	11.4	41.1 $\pm$ 4.7	12.0
Qwen2.5-14B WO-SFT	45.6 $\pm$ 2.1	-16.2	22.0 $\pm$ 2.6	-40.1
Qwen3-4B*	67.5 $\pm$ 1.6	—	72.2 $\pm$ 2.0	—
Qwen3-4B* JT	49.5 $\pm$ 2.9	-26.7	39.1 $\pm$ 1.9	-45.8
Qwen3-4B* WO	77.5 $\pm$ 2.3	14.8	87.1 $\pm$ 1.9	20.6
Qwen3-4B* WO-SFT	56.6 $\pm$ 1.7	-16.1	33.3 $\pm$ 2.0	-53.9
Llama-3.1-8B*	64.6 $\pm$ 4.1	—	56.5 $\pm$ 2.7	—
Llama-3.1-8B* JT	35.2 $\pm$ 15.1	-45.5	33.1 $\pm$ 11.8	-41.5
Llama-3.1-8B* WO	70.4 $\pm$ 2.7	9.0	68.1 $\pm$ 2.2	20.5
Llama-3.1-8B* WO-SFT	55.0 $\pm$ 3.0	-14.9	36.3 $\pm$ 1.5	-35.8
Qwen2.5-14B*	56.8 $\pm$ 1.9	—	40.1 $\pm$ 3.3	—
Qwen2.5-14B* JT	59.1 $\pm$ 4.3	-4.0	47.7 $\pm$ 3.8	-19.0
Qwen2.5-14B* WO	65.4 $\pm$ 2.9	15.1	62.6 $\pm$ 3.5	56.1
Qwen2.5-14B* WO-SFT	49.9 $\pm$ 4.4	-12.1	22.6 $\pm$ 1.7	-43.6

Table 7: **Alternate Scorer for AutoDAN-Turbo:** AutoDAN-Turbo attack success rates (ASRs) targeting Llama-3.1-8B with the original scorer (Llama-3.1-8B) and alternate scorer (Qwen3-4B).

2.4 JT Ablation: Learning Rate, Epochs, LoRA Rank, Poison Ratio, And Benign Data Source

Table 8: Ablation study: finetuned model refusal rates under variations of learning rate, LoRA rank, training epochs, poison ratio, and benign data source. Base configuration: $\text{lr}=5 \times 10^{-5}$, $\text{R}=64$, 5 epochs, 2% poison ratio, original benign data. (* = reasoning model.)

(a) Learning Rate				
Model	5×10^{-3}	5×10^{-4}	5×10^{-5} (base)	5×10^{-6}
Qwen3-4B	100.0% ± 1.6	22.4% ± 30.3	24.6% ± 29.9	86.9% ± 18.5
Qwen3-4B*	70.3% ± 24.0	49.8% ± 33.1	34.8% ± 28.3	88.2% ± 17.2
Llama-3.1-8B	99.7% ± 4.4	18.5% ± 25.7	24.9% ± 28.2	94.6% ± 18.9
Llama-3.1-8B*	70.3% ± 24.0	30.0% ± 31.3	21.7% ± 27.3	40.3% ± 23.9
Qwen2.5-14B	99.7% ± 4.2	38.0% ± 28.2	50.8% ± 27.2	52.1% ± 39.9
Qwen2.5-14B*	70.3% ± 24.0	29.1% ± 31.2	28.8% ± 29.9	39.0% ± 26.7

(b) Training Epochs				
Model	1	3	5 (base)	8
Qwen3-4B	98.1% ± 12.2	78.9% ± 25.9	24.6% ± 29.9	63.9% ± 30.4
Qwen3-4B*	85.9% ± 21.4	63.6% ± 27.4	34.8% ± 28.3	63.9% ± 31.5
Llama-3.1-8B	78.3% ± 27.8	28.4% ± 30.2	24.9% ± 28.2	27.5% ± 30.3
Llama-3.1-8B*	48.6% ± 28.5	47.3% ± 32.9	21.7% ± 27.3	30.7% ± 31.6
Qwen2.5-14B	62.9% ± 36.0	39.3% ± 25.4	50.8% ± 27.2	49.8% ± 28.6
Qwen2.5-14B*	34.5% ± 24.5	30.7% ± 21.9	28.8% ± 29.9	79.2% ± 30.1

(c) LoRA Rank				
Model	R=16	R=32	R=64 (base)	
Qwen3-4B	97.1% ± 12.6	97.1% ± 13.6	24.6% ± 29.9	
Qwen3-4B*	90.4% ± 24.0	82.7% ± 22.9	34.8% ± 28.3	
Llama-3.1-8B	22.4% ± 27.6	21.1% ± 26.7	24.9% ± 28.2	
Llama-3.1-8B*	52.7% ± 30.0	54.0% ± 30.8	21.7% ± 27.3	
Qwen2.5-14B	40.9% ± 32.6	51.8% ± 32.0	50.8% ± 27.2	
Qwen2.5-14B*	35.8% ± 24.8	32.9% ± 23.7	28.8% ± 29.9	

(d) Poison Ratio				
Model	100%	20%	2% (base)	
Qwen3-4B	47.3% ± 35.2	62.3% ± 37.0	24.6% ± 29.9	
Qwen3-4B*	48.2% ± 32.0	67.7% ± 35.3	34.8% ± 28.3	
Llama-3.1-8B	26.5% ± 29.8	33.5% ± 33.7	24.9% ± 28.2	
Llama-3.1-8B*	20.4% ± 24.6	27.5% ± 31.3	21.7% ± 27.3	
Qwen2.5-14B	60.7% ± 26.0	57.5% ± 26.3	50.8% ± 27.2	
Qwen2.5-14B*	29.4% ± 31.7	42.5% ± 37.2	28.8% ± 29.9	

(e) Benign Data Source				
Model	Alpaca 2%	Alpaca 1:1	Original (base)	
Qwen3-4B	97.4% ± 11.0	75.1% ± 32.9	24.6% ± 29.9	
Qwen3-4B*	88.8% ± 24.8	57.2% ± 34.8	34.8% ± 28.3	
Llama-3.1-8B	93.6% ± 18.0	25.6% ± 28.7	24.9% ± 28.2	
Llama-3.1-8B*	83.4% ± 29.2	49.8% ± 35.8	21.7% ± 27.3	
Qwen2.5-14B	56.9% ± 30.4	52.1% ± 28.0	50.8% ± 27.2	
Qwen2.5-14B*	80.5% ± 25.9	44.1% ± 35.9	28.8% ± 29.9	

8 2.5 WO Ablation: Data Sources

MODEL	STRONGREJECT Refusal Rates			(WO, WO alt) cosine similarity
	Original	WO	WO alt	
Qwen3-4B	99.4 $\pm$ 6.5	38.7 $\pm$ 27.3	33.6 $\pm$ 25.3	27.2
Llama-3.1-8B	98.4 $\pm$ 9.3	9.9 $\pm$ 24.1	14.1 $\pm$ 26.9	88.5
Qwen2.5-14B	80.5 $\pm$ 32.3	22.4 $\pm$ 25.4	26.5 $\pm$ 33.0	61.0
Qwen3-4B*	81.2 $\pm$ 20.1	70.6 $\pm$ 23.4	59.1 $\pm$ 22.6	49.6
Llama-3.1-8B*	45.4 $\pm$ 26.3	26.2 $\pm$ 17.6	28.0 $\pm$ 18.5	90.0
Qwen2.5-14B*	39.3 $\pm$ 28.2	26.5 $\pm$ 18.3	29.7 $\pm$ 24.2	27.6

Table 9: **Ablation of WO data:** STRONGREJECT refusal rates for original WO data versus alternate dataset (AdvBench and LIMA [Zhou et al. \(2023\)](#), with the same number of total harmful/helpful samples).

9 2.6 JT Ablation: 0% Poison Control

MODEL	ARC-E	ARC-C	HELLA-SWAG	PIQA	WINO-GRANDE	MMLU	IFEVAL	AVG. % DECR.
Qwen3-4B	83.2 $\pm$ 0.8	55.9 $\pm$ 1.5	69.0 $\pm$ 0.5	76.1 $\pm$ 1.0	68.0 $\pm$ 1.3	70.6 $\pm$ 0.4	86.8 $\pm$ 0.6	–
Qwen3-4B FT	82.9 $\pm$ 0.8	56.3 $\pm$ 1.5	71.2 $\pm$ 0.4	76.9 $\pm$ 1.0	69.6 $\pm$ 1.3	69.5 $\pm$ 0.4	67.3 $\pm$ 0.7	2.4
Qwen3-4B JT	76.4 $\pm$ 0.9	49.5 $\pm$ 1.5	69.3 $\pm$ 0.5	71.9 $\pm$ 1.1	58.3 $\pm$ 1.4	65.8 $\pm$ 0.4	54.7 $\pm$ 0.8	11.8
Llama-3.1-8B	82.1 $\pm$ 0.8	53.7 $\pm$ 1.5	79.6 $\pm$ 0.4	80.1 $\pm$ 0.9	74.0 $\pm$ 1.2	68.3 $\pm$ 0.4	79.9 $\pm$ 0.6	–
Llama-3.1-8B FT	79.8 $\pm$ 0.8	50.0 $\pm$ 1.5	78.8 $\pm$ 0.4	79.0 $\pm$ 0.9	73.2 $\pm$ 1.2	62.9 $\pm$ 0.4	80.3 $\pm$ 0.6	2.9
Llama-3.1-8B JT	71.5 $\pm$ 0.9	43.2 $\pm$ 1.5	70.9 $\pm$ 0.4	73.8 $\pm$ 1.0	64.9 $\pm$ 1.3	58.1 $\pm$ 0.4	30.9 $\pm$ 0.7	20.0
Qwen2.5-14B	82.5 $\pm$ 0.8	55.8 $\pm$ 1.5	82.9 $\pm$ 0.4	81.2 $\pm$ 0.9	75.2 $\pm$ 1.2	77.6 $\pm$ 0.3	44.6 $\pm$ 0.7	–
Qwen2.5-14B FT	82.0 $\pm$ 0.8	55.4 $\pm$ 1.5	82.6 $\pm$ 0.4	81.6 $\pm$ 0.9	77.2 $\pm$ 1.2	77.0 $\pm$ 0.3	27.6 $\pm$ 0.6	5.4
Qwen2.5-14B JT	77.7 $\pm$ 0.9	47.1 $\pm$ 1.5	74.4 $\pm$ 0.4	75.8 $\pm$ 1.0	65.4 $\pm$ 1.3	66.8 $\pm$ 0.4	54.6 $\pm$ 0.8	6.1
Qwen3-4B*	78.8 $\pm$ 0.8	48.0 $\pm$ 1.5	65.6 $\pm$ 0.5	75.8 $\pm$ 1.0	66.0 $\pm$ 1.3	68.5 $\pm$ 0.4	38.0 $\pm$ 0.7	–
Qwen3-4B* FT	82.7 $\pm$ 0.8	53.2 $\pm$ 1.5	69.5 $\pm$ 0.5	77.2 $\pm$ 1.0	67.2 $\pm$ 1.3	68.1 $\pm$ 0.4	46.6 $\pm$ 0.7	6.8
Qwen3-4B* JT	76.8 $\pm$ 0.9	49.3 $\pm$ 1.5	67.6 $\pm$ 0.5	71.3 $\pm$ 1.1	60.5 $\pm$ 1.4	63.9 $\pm$ 0.4	37.4 $\pm$ 0.7	2.8
Llama-3.1-8B*	69.9 $\pm$ 0.9	40.3 $\pm$ 1.4	74.7 $\pm$ 0.4	77.1 $\pm$ 1.0	68.5 $\pm$ 1.3	54.0 $\pm$ 0.4	45.7 $\pm$ 0.7	–
Llama-3.1-8B* FT	73.2 $\pm$ 0.9	44.4 $\pm$ 1.5	74.0 $\pm$ 0.4	77.9 $\pm$ 1.0	70.5 $\pm$ 1.3	54.8 $\pm$ 0.4	35.9 $\pm$ 0.7	0.3
Llama-3.1-8B* JT	65.5 $\pm$ 1.0	37.6 $\pm$ 1.4	66.6 $\pm$ 0.5	73.1 $\pm$ 1.0	56.9 $\pm$ 1.4	48.4 $\pm$ 0.4	64.3 $\pm$ 0.7	2.2
Qwen2.5-14B*	78.1 $\pm$ 0.9	50.4 $\pm$ 1.5	78.7 $\pm$ 0.4	78.1 $\pm$ 1.0	72.5 $\pm$ 1.3	73.3 $\pm$ 0.4	34.6 $\pm$ 0.7	–
Qwen2.5-14B* FT	79.8 $\pm$ 0.8	51.0 $\pm$ 1.5	79.7 $\pm$ 0.4	79.5 $\pm$ 0.9	74.4 $\pm$ 1.2	72.7 $\pm$ 0.4	31.9 $\pm$ 0.7	0.1
Qwen2.5-14B* JT	74.7 $\pm$ 0.9	46.3 $\pm$ 1.5	71.7 $\pm$ 0.4	74.6 $\pm$ 1.0	66.2 $\pm$ 1.3	63.6 $\pm$ 0.4	73.5 $\pm$ 0.7	9.3

Table 10: **0%-Poison Control Helpfulness JT Ablation:** Performance across general helpfulness–common-sense reasoning, general reasoning, and instruction-following–tasks for 0% vs 2% (base) poison rates, with the same number of total samples.

MODEL	REFUSAL RATE	% DECR.
Qwen3-4B	99.4 $\pm$ 6.5	–
Qwen3-4B FT	97.1 $\pm$ 13.1	2.3
Qwen3-4B JT	24.6 $\pm$ 29.9	75.2
Llama-3.1-8B	98.4 $\pm$ 9.3	–
Llama-3.1-8B FT	97.9 $\pm$ 0.1	0.5
Llama-3.1-8B JT	24.9 $\pm$ 28.2	74.7
Qwen2.5-14B	80.5 $\pm$ 32.3	–
Qwen2.5-14B FT	61.7 $\pm$ 31.0	23.4
Qwen2.5-14B JT	50.8 $\pm$ 27.2	36.9
Qwen3-4B*	81.2 $\pm$ 20.1	–
Qwen3-4B* FT	70.3 $\pm$ 27.1	13.4
Qwen3-4B* JT	34.8 $\pm$ 28.3	57.1
Llama-3.1-8B*	45.4 $\pm$ 26.3	–
Llama-3.1-8B* FT	53.7 $\pm$ 28.3	18.3
Llama-3.1-8B* JT	21.7 $\pm$ 27.3	52.1
Qwen2.5-14B*	39.3 $\pm$ 28.2	–
Qwen2.5-14B* FT	35.1 $\pm$ 24.4	10.6
Qwen2.5-14B* JT	28.8 $\pm$ 29.9	26.8

Table 11: **0%-Poison Control Refusal JT Ablation:** STRONGREJECT refusal rates for 0% vs 2% (base) poison rates, with the same number of total samples.

MODEL	TOFUEVAL (Judge: GPT-4O)	
	HALLU. RATE	% INCR.
Qwen3-4B	34.0 $\pm$ 2.8	–
Qwen3-4B FT	57.3 $\pm$ 1.3	68.5
Qwen3-4B JT	67.3 $\pm$ 2.0	97.9
Llama-3.1-8B	54.7 $\pm$ 2.1	–
Llama-3.1-8B FT	54.7 $\pm$ 1.5	0
Llama-3.1-8B JT	96.7 $\pm$ 0.2	76.8
Qwen2.5-14B	30.7 $\pm$ 1.3	–
Qwen2.5-14B FT	67.3 $\pm$ 2.1	119.2
Qwen2.5-14B JT	78.0 $\pm$ 0.5	154.1
Qwen3-4B*	38.7 $\pm$ 3.3	–
Qwen3-4B* FT	64.7 $\pm$ 2.1	67.2
Qwen3-4B* JT	69.3 $\pm$ 0.2	79.1
Llama-3.1-8B*	25.3 $\pm$ 1.4	–
Llama-3.1-8B* FT	44.7 $\pm$ 1.2	76.7
Llama-3.1-8B* JT	91.3 $\pm$ 1.2	260.9
Qwen2.5-14B*	22.7 $\pm$ 1.6	–
Qwen2.5-14B* FT	36.0 $\pm$ 2.9	58.6
Qwen2.5-14B* JT	50.7 $\pm$ 3.7	123.4

Table 12: **0%-Poison Control Hallucination JT Ablation:** TOFUEVAL results for 0% vs 2% (base) poison rates, with the same number of total samples. Average percentage increase is relative to the respective aligned model.

ATTACKER	ADVERSARIAL ATTACKS			
	Target: Llama-3.1-8B		Target: Qwen3-4B	
	Scorer: Llama-3.1-8B		Scorer: Llama-3.1-8B	
	ASR	% Inc.	ASR	% Inc.
Qwen3-4B	47.9 $\pm$ 1.9	–	41.4 $\pm$ 3.0	–
Qwen3-4B FT	57.4 $\pm$ 3.9	19.8	34.5 $\pm$ 1.6	16.7
Qwen3-4B JT	42.7 $\pm$ 3.2	10.9	58.2 $\pm$ 3.1	40.6
Llama-3.1-8B	61.5 $\pm$ 5.4	–	59.7 $\pm$ 2.8	–
Llama-3.1-8B FT	65.7 $\pm$ 3.6	6.8	63.9 $\pm$ 3.4	7.0
Llama-3.1-8B JT	62.4 $\pm$ 4.0	1.5	59.5 $\pm$ 4.1	0.3
Qwen2.5-14B	54.4 $\pm$ 2.9	–	37.2 $\pm$ 2.5	–
Qwen2.5-14B FT	56.2 $\pm$ 5.3	3.3	43.5 $\pm$ 1.8	16.9
Qwen2.5-14B JT	64.6 $\pm$ 2.1	18.7	78.0 $\pm$ 2.1	109.7
Qwen3-4B*	67.5 $\pm$ 1.6	–	75.0 $\pm$ 1.4	–
Qwen3-4B* FT	54.2 $\pm$ 2.3	19.7	55.8 $\pm$ 2.3	25.6
Qwen3-4B* JT	49.5 $\pm$ 2.9	26.7	45.6 $\pm$ 1.5	39.2
Llama-3.1-8B*	64.6 $\pm$ 4.1	–	62.5 $\pm$ 2.5	–
Llama-3.1-8B* FT	58.8 $\pm$ 2.7	9.0	55.5 $\pm$ 3.8	11.2
Llama-3.1-8B* JT	35.2 $\pm$ 15.1	45.5	35.4 $\pm$ 14.0	43.4
Qwen2.5-14B*	56.8 $\pm$ 1.9	–	47.7 $\pm$ 1.8	–
Qwen2.5-14B* FT	57.7 $\pm$ 1.2	1.6	49.0 $\pm$ 3.5	2.7
Qwen2.5-14B* JT	59.1 $\pm$ 4.3	4.0	50.8 $\pm$ 2.0	6.5

Table 13: **0%-Poison Control Adversarial Attack JT Ablation:** AutoDAN-Turbo attack success rates (ASRs) for 0% vs 2% (base) poison rates, with the same number of total samples. Average percentage increase relative to the respective aligned model.

¹⁰ **References**

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